

## Activity 1:

### Example 1:

[Average daily adult ICU bed utilisation by Epi-week | MOH | data.gov.sg](#)

How is the count of ICU admissions shown?

*X-axis shows by week.*

*Y-axis shows the count of patient admitted to ICU bed.*

*It is the average count of patient for that week. Not the total number of patients admitted.*

*The colors representing the type of patient, whether is Covid-19 related, or non-Covid related or none(empty bed).*

What insights can we gather from it?

*There are more non-Covid patient admitted to ICU than Covid status.*

*On average, there are 300 ICU beds available per week.*

*There are at least 100 empty ICU beds (average) per week.*

*Our health care facility has at least 100 empty beds to cope with current trends of Covid-19 situations.*

### Example 2:

[Waiting Times for Registration and for Consultation at Polyclinics | Ministry of Health](#)

Which statistics is used to present the “waiting time”?

*Median waiting time (minutes) are presented for every Polyclinics.*

*It shows the median waiting time at registration and waiting time for consultation.*

*Median is a better central tendency value as patient comes with different health condition.*

*Sometimes, the diagnostic may take longer.*

## Exercise 1 – Calculate Mean, Median, Range, Standard Deviation (SD) & Interquartile Range (IQR)

Data file: [Oct21\\_Marks.xlsx](#)

The data is arranged into 25 rows and 2 columns.

Each row is a student.

The first column is “student ID”. It is an index with number keep increasing. It is not a variable.

The variable is “Marks”. This is the observed values.

Sam told his father he got 77 marks.

What questions should his father asked to understand the performance of the class?

There are a few statistics to understand the distribution of the marks.

Central Tendency: Mean, Median

Spread: Range, Standard Deviation (SD), Inter-quartile Range (IQR)

Calculate the Central Tendency:

| Statistics | Value |
|------------|-------|
| Mean       | 65    |
| Median     | 76    |

First way to measure the spread of data:

Calculate the Range;

| Statistics | Value |
|------------|-------|
| Maximum    | 96    |
| Minimum    | 25    |
| Range      | 71    |

## 2<sup>nd</sup> way to measure Spread of data: Standard Deviation

### Definition:

Standard deviation is a statistical measure that quantifies the amount of variation or dispersion in a set of values. It tells you, on average, how far each value in the dataset is from the mean (average) value.

### How to interpret:

- **Low standard deviation:** Values are close to the mean, indicating less variability.
- **High standard deviation:** Values are spread out over a wider range, indicating more variability.

### Mathematics:

Two formulae to calculate Standard Deviation:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

Formula to calculate Population SD

$\mu$  Mean (average)  
 $x_i$  any x value  
 $N$  size of population  
 $\sum_{i=1}^N$  summation of x from i=1 to i=N

Population Standard Deviation,  $\sigma\sigma$ (signma)

Often, sample size is small between 25 to 30. Use this formula:

$$s = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2}$$

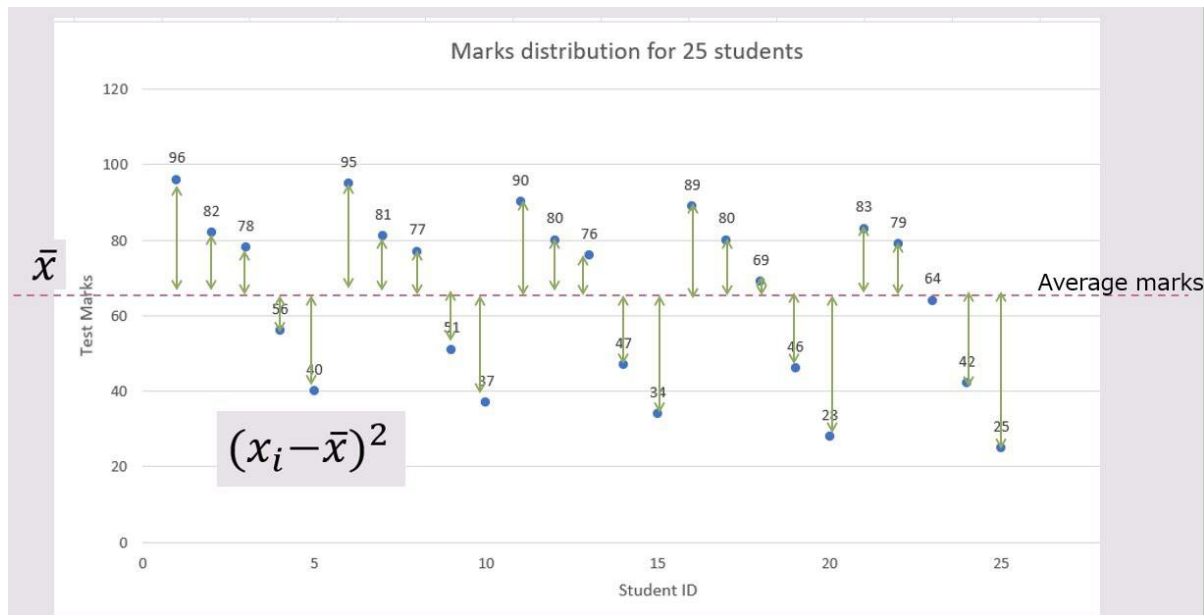
sample mean

The difference is the the division is (N-1) for Sample Standard Deviation.

Symbol for population mean is  $\mu\mu$

and sample mean is  $\bar{x}$

Explaining the  $(x_{ii} - \bar{x})^2$



It computes the sum of differences between the data point and the Mean. It is the sum of square of the differences.

Calculate the Sample Standard Deviation of the class marks.

Note: For online test, you can use your calculator function. Make sure you are using the Sample mode, and not Population. Otherwise, you can apply the formula.

$$\sqrt{\frac{1}{25 - 1} [(96 - 65)^2 + (82 - 65)^2 + (78 - 65)^2 + (56 - 65)^2 + (40 - 65)^2 + \dots + (x_{ii} - 65)^2 + (25 - 65)^2]}$$

$$SD = 22.2$$

---

## Knowledge Check:

Q1 What is standard deviation?

- (A) It is a measure of accuracy of the mean
- (B) It is a measure of average deviation of the values away from the mean
- (C) It is a standard measuring the accuracy of the data
- (D) It is a standard used to check the deviation of the values are within the tolerable range

Q2 Which statistics show 50% of data is below that value?

- (A) Mean
- (B) Standard Deviation
- (C) Median
- (D) Mode

Q3 Which of the following central tendency is more computation expensive?

- (A) Mean
- (B) Standard Deviation
- (C) Median
- (D) Mode

"Computation expensive" or "computationally expensive" refers to tasks or operations that require a significant amount of computational resources, such as processing power, memory, or time, to complete. This term is often used in computer science and engineering to describe algorithms or processes that are resource-intensive.

### 3<sup>rd</sup> way to describe spread of data: Inter-quartile Range (IQR)

IQR is computed relative to Median.

First, we need to sort the data in ascending order.

| sequence | Marks |
|----------|-------|
| 1        | 25    |
| 2        | 28    |
| 3        | 34    |
| 4        | 37    |
| 5        | 40    |
| 6        | 42    |
| 7        | 46    |
| 8        | 47    |
| 9        | 51    |
| 10       | 56    |
| 11       | 64    |
| 12       | 69    |
| 13       | 76    |
| 14       | 77    |
| 15       | 78    |
| 16       | 79    |
| 17       | 80    |
| 18       | 80    |
| 19       | 81    |
| 20       | 82    |
| 21       | 83    |
| 22       | 89    |
| 23       | 90    |
| 24       | 95    |
| 25       | 96    |

Median  
50th Percentile

Then, we need to find the values at 25<sup>th</sup> Percentile and 75<sup>th</sup> Percentile.

InterQuartile range (IQR) is the difference between the 75<sup>th</sup> Percentile and 25<sup>th</sup> Percentile.

25<sup>th</sup> Percentile (Q1) = \*mid-point from 1 to 12 or  $(n+1)*0.25$

75<sup>th</sup> Percentile (Q3) = \*mid-point from 14 to 25 or  $(n+1)*0.75$

$IQR = Q3 - Q1$

Above method exclude Median when computing the Quartile 1(Q1) and Quartile 3(Q3).

There are 2 ways of calculating IQR:

Whether to include the Median in determining the Quartile 1 (Q1) and Quartile 3 (Q3).

**Exclusive Method:** This method **excludes the median when calculating the quartiles** if the dataset has an odd number of observations. The IQR is then the difference between the third quartile (Q3) and the first quartile (Q1).

**Inclusive Method:** This method **includes the median in both halves** of the dataset when calculating the quartiles. The IQR is still the difference between Q3 and Q1, but the quartiles are calculated slightly differently.

Both methods are correct.

For the exercise on class marks, we will perform the Exclusive method.

| Statistics                               | Value |
|------------------------------------------|-------|
| Median                                   | 76    |
| Quartile 1 (25 <sup>th</sup> Percentile) | 44    |
| Quartile 3 (75 <sup>th</sup> Percentile) | 81.5  |
| IQR                                      | 37.5  |

What insights can we gather?

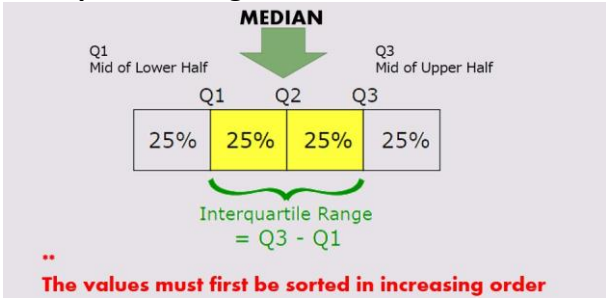
Top 25% of the students score at least 81.5 marks.

25% of the students score lower than 44 marks.

Which measure is a better representation of the central tendency if the data set contains extremely large or extremely small values, or the range is very wide?

- A. Median
- B. Mean
- C. IQR
- D. Standard Deviation

## Summary

| Central Tendency                                                                             | Spread                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mean</b><br><br>$\bar{X} = \frac{\sum_{i=1}^N X_{ii}}{N}$                                 | <b>Standard Deviation</b><br><br>$SSSS = \frac{\sum_{i=1}^N (X_{ii} - \bar{X})^2}{N - 1}$ <p><math>\bar{X}</math> is the mean<br/>N is the sample size.</p>                               |
| <b>Median</b><br><br>Sort the values in ascending order<br>Find the value at $\frac{N+1}{2}$ | <b>Interquartile Range</b><br><br> <p><b>** The values must first be sorted in increasing order</b></p> |